

Week 8 Office Hour Notes

8.0 Errata

10/22/24 4:00pm. Lectured on Sections 8.2 and 8.3.

10/22/24 8:30pm. Went over the Homework 7 Solutions.

8.1 Homework 7 Solutions

Omitted to not reveal homework solutions...

8.2 Proper First Order Language and Structural Induction on Wff's

In the so-called "proper first order language" (the technical language that we define in the book and powerpoints), we use the following symbols (and all others that you may see appear are shortcut symbols to talk about the others).

Syntax symbols $(,)$

Sentential Connective symbols $\neg, \rightarrow$

Quantifier: $\forall$

Variables: $\{v_n\}_{n \in \mathbb{N}}$

Function Symbols: $\bigcup_{n \in \mathbb{N} \cup \{0\}} \mathcal{F}_n$

Relation/Predicate Symbols: $\{ = \} \cup \bigcup_{n \in \mathbb{N} \cup \{0\}} \mathcal{R}_n$ (unless the particular language in question specifies that it does not contain equality)

So with this construction in mind, note that a wff has the following structural induction property:

Structural Induction on Proper First Order Well Formed Formulas. Suppose any set of wff's S that has the following properties:

(i) (Base case) Contains all atomic formulas (which consists of the base case, i.e. strings of the form $= t_1 t_2$ for terms t_1, t_2 and $R_n t_1 \cdots t_n$ for any n -ary relation symbol R_n and terms $t_1, \dots, t_n$).

(ii) (Inductive Step) If $\phi, \psi \in S$, then we have $(\neg\phi), (\phi \rightarrow \psi) \in S$. and for all $n \in \mathbb{N}$, we have $(\forall v_n \phi) \in S$.

Then S consists of all wff's.

Note that we can talk about the other connectives and quantifiers in terms of the connectives $\neg, \rightarrow$ and quantifier $\forall$ using the following shorthand translation for any wff ϕ, ψ

$\phi \vee \psi$ translates to $\neg\phi \rightarrow \psi$

$\phi \wedge \psi$ translates to $\neg(\phi \rightarrow \neg\psi)$

$\phi \leftrightarrow \psi$ translates to $\neg((\phi \rightarrow \psi) \rightarrow \neg(\psi \rightarrow \phi))$

$\exists x \phi$ translates to $\neg \forall x \neg \phi$

How these translations correspond to the ideal properties of these connectives/quantifiers we'll talk about during week 10.

8.3 Formula Construction Trees

Similar to construction trees on sentential formulas, we have construction trees on first order formulas that link from the atomic sub-formulas to the subformulas that connect each other through the connectives and quantifiers to make the formula as a whole.

As an example, let's take $\forall v_1 \forall v_2 (= v_1 v_2 \rightarrow \forall v_3 (\in v_3 v_1 \rightarrow \in v_3 v_2))$ from Example 7.3.4 (ii). Noting that it's wff created by the following set-builder composition:

$$\mathcal{E}_{\forall v_1}(\mathcal{E}_{\forall v_2}(\mathcal{E}_{\rightarrow} (= v_1 v_2, \mathcal{E}_{\forall v_3}(\mathcal{E}_{\rightarrow} (\in v_3 v_1, \in v_3 v_2)))),$$

we then have the following construction tree:

$$\begin{array}{c} \forall v_1 \forall v_2 (= v_1 v_2 \rightarrow \forall v_3 (\in v_3 v_1 \rightarrow \in v_3 v_2)) \\ \downarrow \forall v_1 \\ \forall v_2 (= v_1 v_2 \rightarrow \forall v_3 (\in v_3 v_1 \rightarrow \in v_3 v_2)) \end{array}$$

$$\begin{array}{c}
\downarrow \forall v_2 \\
(= v_1 v_2 \rightarrow \forall v_3 (\in v_3 v_1 \rightarrow \in v_3 v_2)) \\
\rightarrow \\
\swarrow \quad \searrow \\
= v_1 v_2 \quad \forall v_3 (\in v_3 v_1 \rightarrow \in v_3 v_2) \\
\downarrow \forall v_3 \\
\in v_3 v_1 \rightarrow \in v_3 v_2 \\
\rightarrow \\
\swarrow \quad \searrow \\
\in v_3 v_1 \quad \in v_3 v_2
\end{array}$$

8.4 Questions

Evelin asks the general question of what the difference between writing an "understandable" formula and a formula in so-called "correct first order language". In this section I will answer this question and give you some guidelines of how to do Homework 8 in the process.

Problem 3 of Homework 8 asks to create an "understandable" formula that says "there are at least n (distinct) things". Contrast that with *Problem 2 of Homework 7*, which asks to create a formula in "correct first order language" (in the language of set theory). The general question is what the difference is between when the homework asks to write an "understandable" formula vs. a formula in "correct first order language".

Note that the language in Homework 8 that we're using only has $=$ as a relation symbol and no function symbols.

What we mean by "understandable" is a formula (in a given language) that utilizes only those symbols in the language.

For example, let's say we wanted to write the formula that

"there are n things that are all equal to each other".

We would go about this by calling for those n things, each as a variable $v_1, \dots, v_n$, and describing them through the formulas in the language in a way that equivalently describes what the formula says in English, so we would write the following in the "understandable" translation:

$$\exists v_1 \exists v_2 \cdots \exists v_n [(v_1 = v_2 \wedge v_1 = v_3 \wedge \cdots \wedge v_1 = v_n)]$$

$$\begin{aligned} & \wedge (v_2 = v_1 \wedge v_2 = v_3 \wedge \cdots \wedge v_2 = v_n) \\ & \wedge \cdots \wedge (v_n = v_1 \wedge v_n = v_2 \wedge \cdots \wedge v_n = v_{n-1})]. \end{aligned} \quad (8.5)$$

Note that we can also use the following notation:

$\bigwedge_{j=1}^k \phi_j$ to define $\phi_1 \wedge \phi_2 \wedge \cdots \wedge \phi_n$ (written conceptually in the Polish notation translation $\underbrace{\wedge \cdots \wedge}_{n \text{ times}} \phi_1 \cdots \phi_n$ to avoid confusion with parentheses) for first order wff's $\phi_1, \dots, \phi_n$. We

can make easier to look at by writing (8.5) as

$$\exists v_1 \exists v_2 \cdots \exists v_n \left(\bigwedge_{j=1}^n \left[\bigwedge_{k \in \{1, \dots, n\} \setminus \{j\}} v_j = v_k \right] \right).$$

The difference between "understandable" and "proper first order notation" is

"understandable" can use any connectives and quantifiers plus any of the variables, function, and relation symbols specific to the language, whereas proper first order notation uses only the connectives defined in the language in Enderton/the powerpoints using only connectives $\neg$, $\rightarrow$ and the quantifier $\forall$. So for example, with the sentence

"there exists two equal objects",

$\exists v_1 \exists v_2 (v_1 = v_2)$ would be an example of an "understandable" first order formula translation, but not in "proper first order notation" (since we're using existential quantifiers and not universal quantifiers).

In "proper first order notation", we would use the formula

$$\neg(\forall v_1 \neg(\neg \forall v_2 \neg(v_1 = v_2))).$$

Also another example would be

$$\neg(\forall v_1 \forall v_2 \neg(v_1 = v_2)),$$

which would read

"it's not the case that every object is not equal",

if you interpret the english of that statement (and think about it), it would equate to the existence of two objects that are equal, since the opposite of every object being unequal is the existence of two objects being equal.

Recall that if you have connectives like $\wedge$, $\vee$, $\leftrightarrow$ that are not part of proper first order notation, then we would modify the formula as so to make it proper:

$\phi \wedge \psi$ would modify to $\neg(\phi \rightarrow \neg\psi)$

$\phi \vee \psi$ would modify to $\neg\phi \rightarrow \psi$

$\phi \leftrightarrow \psi$ would modify to $\neg(\phi \rightarrow \psi \rightarrow \neg(\psi \rightarrow \phi))$ (since recall that $\phi \leftrightarrow \psi$ is logically equivalent to $(\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)$)